

Methods: Kinetic Segregation Model of TCR–CD45 Spatial Exclusion on a 2D Membrane Patch

Based on Neve-Oz, Sherman & Raveh, *Frontiers in Immunology*, 2024

1 Model overview

The kinetic segregation (KS) model simulates the spatial segregation of T-cell receptor (TCR) and CD45 molecules on a $2\ \mu\text{m} \times 2\ \mu\text{m}$ membrane patch. A 2D membrane height field $h(x, y)$ is discretised on a regular grid of spacing $\Delta x = L/N_{\text{grid}}$, where $L = 2000\ \text{nm}$ and N_{grid} is typically 64. Static pMHC ligands (N_{pMHC} , default 200) are placed on the opposing antigen-presenting cell (APC) surface and gate TCR–pMHC binding: TCR molecules (N_{TCR} , default 50) are attracted to tight-contact regions (low h) *only in grid cells that contain at least one pMHC*. CD45 molecules (N_{CD45} , default 100) are sterically excluded from regions where h is smaller than the CD45 ectodomain height $h_{\text{CD45}} = 50\ \text{nm}$.

The system is evolved with a Metropolis–Hastings Monte Carlo (MC) algorithm at thermal energy $k_B T = 1$ (all energies are expressed in units of $k_B T$).

2 System energy

The total energy of the system is the sum of three contributions:

$$E_{\text{total}} = E_{\text{bend}}[h] + \sum_{i=1}^{N_{\text{TCR}}} \mathbb{K}_{\text{pMHC}}(\mathbf{r}_i) E_{\text{TCR}}(h(\mathbf{r}_i)) + \sum_{j=1}^{N_{\text{CD45}}} E_{\text{CD45}}(h(\mathbf{r}_j)) \quad (1)$$

where $\mathbf{r}_i$ and $\mathbf{r}_j$ are the lateral positions of individual TCR and CD45 molecules, respectively, $h(\mathbf{r})$ is the membrane height at that position (obtained by nearest-grid-point interpolation), and $\mathbb{K}_{\text{pMHC}}(\mathbf{r})$ is unity when the grid cell at $\mathbf{r}$ contains at least one pMHC and zero otherwise.

3 Potential energy terms

Table 1 summarises all energy contributions, their functional forms, and default parameter values.

3.1 Membrane bending energy

The bending energy follows the Helfrich Hamiltonian discretised on a square grid with periodic boundary conditions. The discrete Laplacian at grid point (i, j) is computed via the standard five-point stencil:

$$\nabla_d^2 h_{ij} = \frac{h_{i-1,j} + h_{i+1,j} + h_{i,j-1} + h_{i,j+1} - 4 h_{ij}}{\Delta x^2} \quad (2)$$

with indices taken modulo N_{grid} (periodic boundaries). The full bending energy is then:

$$E_{\text{bend}} = \frac{\kappa}{2} \sum_{i=0}^{N_{\text{grid}}-1} \sum_{j=0}^{N_{\text{grid}}-1} (\nabla_d^2 h_{ij})^2 \Delta x^2 \quad (3)$$

Table 1: Energy terms in the kinetic segregation model. All energies are in units of $k_B T$; lengths are in nm.

Term	Expression	Parameters	Description
Membrane bending	$E_{\text{bend}} = \frac{\kappa}{2} \sum_{i,j} (\nabla_d^2 h_{ij})^2 \Delta x^2$	κ : bending rigidity ($k_B T$); $\Delta x = L/N_{\text{grid}}$	Helfrich bending energy on a discrete grid with periodic BCs
TCR-pMHC binding	$E_{\text{TCR}}(h)$ $-U_{\text{assoc}} \exp\left(-\frac{(h - h_0^{\text{TCR}})^2}{2\sigma_{\text{bind}}^2}\right)$	$= U_{\text{assoc}} = 20 k_B T$; $\sigma_{\text{bind}} = 3 \text{ nm}$	Gaussian attractive well centred at the bond length h_0^{TCR}
CD45 steric repulsion	$E_{\text{CD45}}(h)$ $\begin{cases} \frac{1}{2} k_{\text{rep}} (h_{\text{CD45}} - h)^2 & h < h_{\text{CD45}} \\ 0 & h \geq h_{\text{CD45}} \end{cases}$	$= k_{\text{rep}} = 1.0 k_B T/\text{nm}^2$; $h_{\text{CD45}} = 50 \text{ nm}$	Harmonic repulsive wall; excludes CD45 from tight contact

3.2 TCR-pMHC binding potential

TCR molecules are attracted to regions of low membrane height (tight contact between the T-cell and antigen-presenting cell), modelling TCR-pMHC engagement. The potential is a Gaussian well centred at the resting TCR-pMHC bond length $h_0^{\text{TCR}} = 13 \text{ nm}$ —*not* at $h = 0$. A bound TCR is equally disfavoured when the membranes are too far apart and when they are squeezed closer than the bond allows:

$$E_{\text{TCR}}(h) = -U_{\text{assoc}} \exp\left(-\frac{(h - h_0^{\text{TCR}})^2}{2\sigma_{\text{bind}}^2}\right) \quad (4)$$

where $U_{\text{assoc}} = 20 k_B T$ is the association energy depth and $\sigma_{\text{bind}} = 3 \text{ nm}$ is the spatial width of the binding well.

3.3 CD45 steric repulsion

CD45 has a large extracellular domain ($\sim 35 \text{ nm}$) and is sterically excluded from tight-contact zones. This is modelled as a one-sided harmonic potential:

$$E_{\text{CD45}}(h) = \begin{cases} \frac{1}{2} k_{\text{rep}} (h_{\text{CD45}} - h)^2 & \text{if } h < h_{\text{CD45}} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

with $k_{\text{rep}} = 1.0 k_B T/\text{nm}^2$ and $h_{\text{CD45}} = 50 \text{ nm}$.

4 Monte Carlo algorithm

The simulation uses a Metropolis-Hastings MC scheme with two alternating update phases per sweep. Each *sweep* updates every degree of freedom once (all molecules and all grid cells), ensuring ergodic sampling of the configuration space.

Algorithm 1 gives the complete pseudocode.

4.1 Step sizes

Step sizes are derived from Brownian dynamics. A stability-limited time step Δt is computed as

$$\Delta t = \frac{\Delta x^2}{2 D_h \kappa} \alpha_{\text{safety}} \quad (6)$$

with $\alpha_{\text{safety}} = 0.05$ (MAX_BENDING_DE_PER_STEP), the maximum typical bending ΔE per grid point per step. A second reduction then applies: $\Delta t \leftarrow \min(\Delta t, \ell^2/(8D_{\text{mol}}))$, where ℓ is the smallest resolved length among Δx , σ_r (gaussian binding with pMHC present) and the molecular repulsion cutoff. This keeps a proposed lateral move small relative to every feature it must resolve. The molecular and membrane height step sizes are then

$$\sigma_{\text{step}} = \sqrt{2 D_{\text{mol}} \Delta t}, \quad \sigma_h = \sqrt{2 D_h \Delta t} \quad (7)$$

where $D_{\text{mol}} = 10^4 \text{ nm}^2/\text{s}$ and $D_h = 5 \times 10^4 \text{ nm}^2/\text{s}$ are the molecular and membrane height diffusion coefficients. An alternative “paper” mode uses fixed $\Delta t = 0.01 \text{ s}$ and $\sigma_h = 1 \text{ nm}$.

4.2 Efficient bending energy update

Rather than recomputing the full bending energy (an $O(N_{\text{grid}}^2)$ operation), the change in bending energy when a single grid cell (i, j) is updated is computed in $O(1)$ time. Only the five Laplacian stencil cells affected by the change— (i, j) itself and its four periodic neighbours—need to be considered:

$$\Delta E_{\text{bend}} = \frac{\kappa}{2} \Delta x^2 \sum_{(a,b) \in \mathcal{N}(i,j)} \left[(\nabla_d^2 h_{ab}^{\text{new}})^2 - (\nabla_d^2 h_{ab}^{\text{old}})^2 \right] \quad (8)$$

where $\mathcal{N}(i, j) = \{(i, j), (i \pm 1, j), (i, j \pm 1)\}$ with periodic wrapping.

4.3 Checkerboard decomposition for Phase 2

Phase 2 (membrane height updates) uses a red–black checkerboard decomposition. Grid cells are coloured so that each cell of a given colour has only opposite-colour neighbours in the five-point stencil. All same-colour cells can then be updated simultaneously without data races, because their energy deltas depend only on the frozen opposite-colour neighbours.

Each sweep processes both colours sequentially (red then black), ensuring that every grid cell is updated exactly once. On the CPU this proceeds serially within each colour; on the GPU, all cells of a given colour are dispatched in parallel.

4.4 Initial conditions

- The membrane height field is initialised uniformly at $h = h_{\text{init}} = 70 \text{ nm}$, with a central disc (radius $\sim N_{\text{grid}}/8$ cells) depressed to $h_0^{\text{TCR}} = 13 \text{ nm}$ to seed a tight-contact zone.
- pMHC ligands are placed on the APC surface—either uniformly across the patch, or within a centred disc of radius $L/3$ (“inner-circle” mode). Uniform placement is the default; note that inner-circle is required for the centre-referenced depletion metrics to be meaningful. They remain static throughout the simulation.
- TCR molecules are co-located with randomly chosen pMHC positions, with a small Gaussian jitter (σ_{bind}), so that each TCR starts near a cognate pMHC. When no pMHC are present, TCRs are placed with a Gaussian bias towards the patch centre (spread $L/6$).
- CD45 molecules are placed uniformly at random across the patch.

4.5 Output

The primary observable is the *depletion zone width*, defined as the difference between the median radial distance from the patch centre for CD45 molecules and that for TCR molecules:

$$w_{\text{depletion}} = \max\left(0, \tilde{r}_{\text{CD45}} - \tilde{r}_{\text{TCR}}\right) \quad (9)$$

where $\tilde{r}$ denotes the median over all molecules of that species. The use of the median rather than a percentile-based measure provides robustness at small molecule counts.

5 Simulation parameters

Table 2 lists all tunable parameters and their default values from Supplementary Table S1 of the reference.

Table 2: Default simulation parameters.

Parameter	Symbol	Default value
Patch size	L	2000 nm (2 μm)
Grid resolution	N_{grid}	64
Grid spacing	Δx	$L/N_{\text{grid}} \approx 31.25$ nm (but see §5.1)
Number of TCR	N_{TCR}	125
Number of CD45	N_{CD45}	500
Number of pMHC	N_{pMHC}	auto: 300/ μm^2 over the placement area
Association energy	U_{assoc}	20 $k_B T$
Binding well width	σ_{bind}	3 nm
CD45 ectodomain	h_{CD45}	50 nm
Repulsive constant	k_{rep}	1.0 $k_B T/\text{nm}^2$
Initial height	h_{init}	70 nm
TCR contact height	h_0^{TCR}	13 nm
Mol. diffusion	D_{mol}	10 ⁴ nm ² /s
Height diffusion	D_h	5 $\times$ 10 ⁴ nm ² /s
Safety factor	α_{safety}	0.05
Bending rigidity	κ	Swept (input parameter, $k_B T$)
Simulation time	t	Swept (input parameter)
Random seed	—	42 (per-point FNV-1a hash derived)

5.1 Grid resolution and the pMHC influence field

In **gaussian** binding mode the TCR well at each cell is scaled by a pMHC influence weight $W = \min(1, \sum_k \exp(-r_k^2/2\sigma_r^2))$, summed over pMHC within $3\sigma_r$ of the cell centre, with $\sigma_r = 2$ nm by default.

This makes Δx a physical parameter rather than a discretisation choice. When $\Delta x \gg \sigma_r$ no cell centre lies within the cutoff, W is identically zero, and the attractive term vanishes—so a seeded contact simply relaxes to h_{init} . Measured at fixed physical time, the mean height inside the contact disc ends at 63.6 nm for $\Delta x = 41.7$ nm versus 20.2 nm for $\Delta x = 7.8$ nm. Configurations intended to exercise binding should therefore keep Δx within roughly a factor of a few of σ_r ; the behaviour sweep in **experiments/ks_behavior_sweep/** uses $L = 500$ nm with $N_{\text{grid}} = 100$ ($\Delta x = 5$ nm) for this reason.

6 Implementation

The model is implemented as a portable C99 simulation core with a C++20 command-line interface and an optional GPU backend. A single binary (**ks_gpu**) provides both CPU and GPU execution modes. The build system uses CMake with platform detection: Apple Metal on macOS, with a planned CUDA backend for Linux.

6.1 Code architecture

The source is organised into four layers (Table 3).

Table 3: Source code organisation.

Layer	Language	Files / purpose
Simulation core	C99	<code>simulation.c</code> , <code>potentials.c</code> , <code>rng.c</code> — MC sweep, molecular potentials, PCG64 RNG. Fully portable; no platform dependencies.
Shared physics	C99 / MSL	<code>ks_physics.h</code> — shared <code>static inline</code> float functions for TCR binding, CD45 repulsion, Laplacian, and bending-energy delta. Preprocessor guards (<code>__METAL_VERSION__</code> , <code>__CUDA_ARCH__</code>) select Metal address-space qualifiers or CUDA decorators.
GPU backend	Objective-C / MSL	<code>gpu_engine.h</code> (backend-neutral C API), <code>metal_engine.m</code> (Metal implementation), <code>shaders.metal</code> (GPU compute kernels).
CLI	C++20	<code>main.cpp</code> — argument parsing, JSON I/O (nlohmann/json, header-only), directory management (<code>std::filesystem</code>), deterministic seed derivation (FNV-1a hash). No platform-specific dependencies.

The shared physics header (`ks_physics.h`) is the single source of truth for the four float-precision energy functions used by both the CPU Phase 2 path (`simulation.c`) and the GPU evaluate kernel (`shaders.metal`). This eliminates the previous code duplication and guarantees CPU–GPU equivalence by construction.

6.2 GPU backend interface

The GPU backend is accessed through a four-function C API defined in `gpu_engine.h`: `gpu_engine_create`, `gpu_engine_destroy`, `gpu_engine_h_ptr`, and `gpu_engine_grid_update`. The interface uses an opaque `void*` handle, allowing different backends (Metal, CUDA) to be linked at compile time without changing the simulation core.

6.3 Backend selection and reporting

The GPU path is used by default when it is available, and `-no-gpu` forces the CPU path. The two are not numerically identical: the GPU evaluates the energies in single precision and draws from a Philox counter-based stream, while the CPU path uses double precision and PCG64, so the two are statistically equivalent realisations rather than the same trajectory. Reported results must therefore state which path produced them, and every run records the backend both on `stderr` and as `diagnostics.backend` ("metal" or "cpu") in the output JSON.

Availability is determined solely by the flags and by whether a Metal device and shader are present. The shader is located relative to the *executable*, never the working directory: until 2026-08-11 the lookup was working-directory relative, which made the backend — and hence the numerical result — depend on the directory the caller happened to invoke the binary from. Where a run must not silently degrade, `-require-gpu` makes an unavailable GPU backend a fatal error instead of a fallback.

6.4 CPU mode

In CPU mode, Phase 1 (molecular moves) and Phase 2 (grid height updates) both execute sequentially. Random numbers are generated with the PCG64 algorithm, seeded deterministically from the user-provided seed. Phase 2 uses the checkerboard decomposition described above but updates each colour’s cells serially.

6.5 GPU acceleration (Metal)

On Apple Silicon, Phase 2 is offloaded to the GPU via Metal compute shaders. The pipeline consists of four kernel dispatches per colour:

1. **Propose:** Each thread proposes a new height for one cell using a Philox counter-based RNG.
2. **Snapshot:** Copies the full height field to a frozen buffer so that the evaluate kernel reads consistent neighbour values.
3. **Evaluate:** Computes the energy delta (bending + molecular contributions) and the Metropolis acceptance criterion.
4. **Apply:** Writes accepted proposals back to the authoritative height buffer.

Phase 1 remains on the CPU (few molecules, inherently sequential). The height field resides in a Metal shared buffer (`MTLResourceStorageModeShared`), allowing zero-copy access from both CPU and GPU on unified memory.

A grid-substep parameter K allows batching K Phase 2 sweeps per Phase 1 molecular move, reducing CPU–GPU synchronisation overhead. All $K \times 8$ kernel dispatches are encoded into a single Metal command buffer before committing.

6.6 Performance

Table 4 shows representative timings on Apple M3 for 500-step simulations.

Table 4: CPU vs GPU wall-clock time (500 steps, $\kappa = 20$, $N_{\text{TCR}} = 125$, $N_{\text{CD45}} = 500$).

N_{grid}	CPU (s)	GPU (s)	Speedup
64	0.14	0.06	$2.3\times$
128	0.50	0.06	$8.7\times$
256	1.78	0.07	$25\times$
512	6.29	0.09	$69\times$

References

Y. Neve-Oz, E. Sherman, and B. Raveh. Bayesian metamodeling of early T-cell antigen receptor signaling accounts for its nanoscale activation patterns. *Frontiers in Immunology*, 2024.

Algorithm 1 Metropolis Monte Carlo for kinetic segregation

Require: N_{steps} , membrane grid h , positions $\{\mathbf{r}_i^{\text{TCR}}\}$, $\{\mathbf{r}_j^{\text{CD45}}\}$, parameters $\kappa, U_{\text{assoc}}, \sigma_{\text{bind}}, h_{\text{CD45}}, k_{\text{rep}}$

Ensure: Final configuration $\{h, \mathbf{r}^{\text{TCR}}, \mathbf{r}^{\text{CD45}}\}$

1: **for** $s = 1$ **to** N_{steps} **do**

▷ **Phase 1: Molecular moves**

2: **for** each molecule k in $\text{TCR} \cup \text{CD45}$ **do**

3: $\mathbf{r}_{\text{old}} \leftarrow \mathbf{r}_k$

4: $E_{\text{old}} \leftarrow E_{\text{mol}}(\mathbf{r}_k, h)$

▷ Eq. 4 or 5

5: $\mathbf{r}_k \leftarrow \mathbf{r}_k + \mathcal{N}(0, \sigma_{\text{step}}^2) \hat{\mathbf{e}}$

▷ Gaussian displacement in x, y

6: Wrap $\mathbf{r}_k$ with periodic boundaries (mod L)

7: $E_{\text{new}} \leftarrow E_{\text{mol}}(\mathbf{r}_k, h)$

8: $\Delta E \leftarrow E_{\text{new}} - E_{\text{old}}$

9: **if** $\Delta E \leq 0$ **or** $\log \mathcal{U}(0, 1) < -\Delta E$ **then**

10: Accept move

11: **else**

12: $\mathbf{r}_k \leftarrow \mathbf{r}_{\text{old}}$

▷ Reject

13: **end if**

14: **end for**

▷ **Phase 2: Membrane height updates**

15: **for** each grid cell (i, j) **do**

16: $h_{\text{old}} \leftarrow h_{ij}$

17: $E_{\text{mol,old}} \leftarrow$ sum of E_{mol} for all molecules in cell (i, j)

18: $h_{ij} \leftarrow h_{\text{old}} + \mathcal{N}(0, \sigma_h^2)$; **if** $h_{ij} < 0$ **then** $h_{ij} \leftarrow -h_{ij}$

▷ Reflect to keep $h \geq 0$

19: $\Delta E_{\text{bend}} \leftarrow$ local bending energy change (5-point stencil)

▷ $O(1)$ via Eq. 2

20: $E_{\text{mol,new}} \leftarrow$ sum of E_{mol} for all molecules in cell (i, j) at new h_{ij}

21: $\Delta E \leftarrow \Delta E_{\text{bend}} + (E_{\text{mol,new}} - E_{\text{mol,old}})$

22: **if** $\Delta E \leq 0$ **or** $\log \mathcal{U}(0, 1) < -\Delta E$ **then**

23: Accept move

24: **else**

25: $h_{ij} \leftarrow h_{\text{old}}$

▷ Reject

26: **end if**

27: **end for**

28: **end for**

29: **Output:** Depletion width $w = \text{median}(r^{\text{CD45}}) - \text{median}(r^{\text{TCR}})$
